

Searching Files on Your System

For *messier users*, or *users with a LOT of data* to process; trying to find a file where you have the idea in your head, but you **DO NOT remember the name** is a situation all too common.

- This is *EASILY remedied*, by searching with the **TERMINAL**
-

Let's **MAKE** some *Mock Data* for **SEARCHING**:

- We're going to make a **NEW** folder for *our searching lab*.
 - [Mock Data - Pastebin.com](#) This has the **MOCK** data command, it'll make a and populate a **FOLDER** called **search_lab**. *Right click* to paste into the **TERMINAL**.

```
vboxuser@ExampleName:~/search_lab$ ls
CHANGELOG.txt  Downloads  logs      photos    README.txt
Documents      LICENSE.md misc       projects  work
```

- There is a LOT of data, but we have a *handy tool* to find our stuff.
-

> **find** is a command to **evidently** **FIND** stuff, *given filters*.

Our First Command

> **find ./ -name "reports"** This will look; **FROM** your current directory (**./**) for *anything* with the name (**-name**), **"reports"** in its **FILE NAME**.

```
vboxuser@ExampleName:~/search_lab$ find ./ -name "reports"
./work/reports
```

We only got the *EXACT match* of the name no **"reports2024"** or anything. Let's allow **LOOSE matches**. *Plus*, this is a **directory**, we *only want files*.

> **find ./ -type f -name "*report*"**

- Yet again, **FROM** your current folder (**./**) look for *anything* that is a **FILE** (**-type f**). For anything that **loosely** matches the name (**-name**) **"report"**, the ****** around the word **"reports"**, means look for the word **"report"**, but *IF* (or not) there's anything on the **LEFT** or **RIGHT** allow that too.

We can use this * a lot, FILE TYPES are PART of the name, so we can SEARCH for that name. (We don't need one on the right as there's nothing after)

> `find ./ -name "*.pdf"` Find anything from here, but it MUST end with .pdf

```
vboxuser@ExampleName:~/search_lab$ find ./ -name "*.pdf"
./Documents/Important/passport_scan.pdf
./projects/client_b/presentations/product_demo.pdf
./projects/client_a/docs/contract.pdf
./work/presentations/client_deck.pdf
./work/meetings/agenda.pdf
./work/reports/2024/q2_report.pdf
```

There is ANOTHER type of special character.

> `find ./ -name "file?.pdf"` This finds any FILE that is named file.pdf, but leaves space for ONLY ONE uncertain character after the word file represented by the ?. Meaning, `file1.pdf`, `filex.pdf`, BUT NOT `file.pdf` outputs.

- This is different, as the (*) accepts any amount of uncertain characters.

```
vboxuser@ExampleName:~/search_lab$ find ./ -name "file?.pdf"
./file1.pdf
./filex.pdf
```

Currently, these files have NO DATA. For the now we'll have files with CONTENT so you can search by the metadata of the files.

The LAST **paste** had *no data inside*, so we'll use **this new mock dataset**.

(You can delete the old one `rm -rf search_lab`)

From your home (~), *right click* and **PASTE**. Creates data with INFO to search [Finding IN Files - Pastebin.com](#)

Let's **find** FILES that are *MORE THAN 1 Megabyte* for CLEANING up our storage.

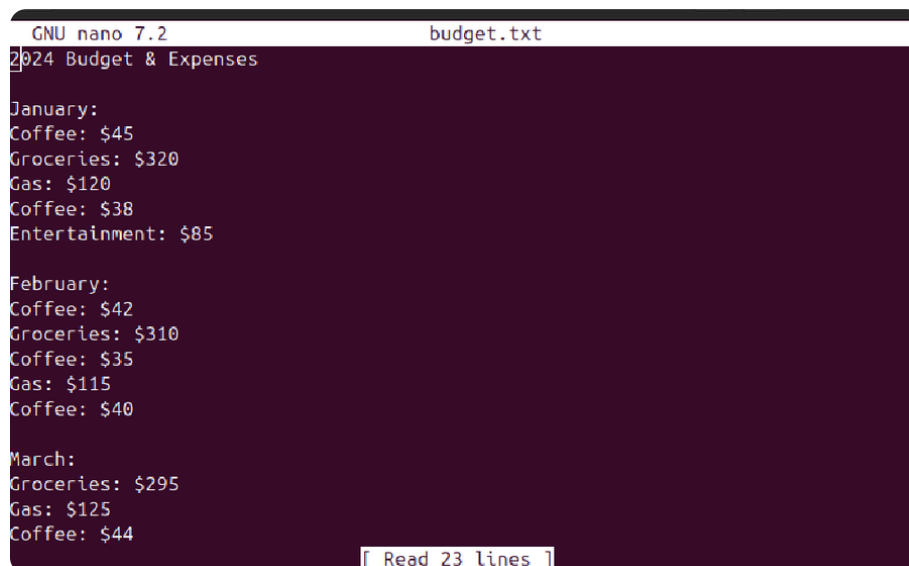
> `find ./ -type f -size +1M` Looks for ANY file with a size of *MORE (+) than 1MB*

K → KB, M → MB, G → GB

```
vboxuser@ExampleName:~/search_practice$ find ./ -size +1M
./old_vacation_videos.mp4
./cache_files.tmp
```

Let's open a file real quick, **nano** is a tool to open edit text files

> `nano budget.txt`



```
GNU nano 7.2 budget.txt
2024 Budget & Expenses

January:
Coffee: $45
Groceries: $320
Gas: $120
Coffee: $38
Entertainment: $85

February:
Coffee: $42
Groceries: $310
Coffee: $35
Gas: $115
Coffee: $40

March:
Groceries: $295
Gas: $125
Coffee: $44

[ Read 23 lines ]
```

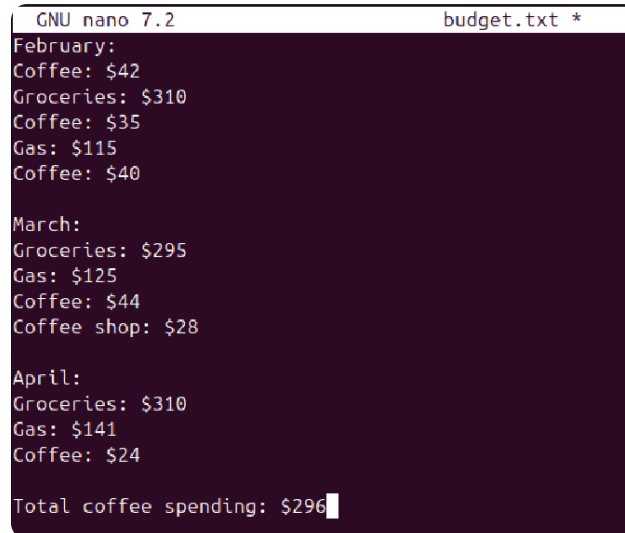
(CTRL+X to exit, don't make any changes)

NOW, let's RUN OUR COMMAND

> `find ./ -atime 0` This means; from the current DIRECTORY, find anything that has been **a**ccessed in the last **0** days (today)

```
vboxuser@ExampleName:~/search_practice$ find ./ -atime 0
./
./budget.txt
```

Now let's EDIT a file; nano budget.txt and let's add a new line



```
GNU nano 7.2 budget.txt *
February:
Coffee: $42
Groceries: $310
Coffee: $35
Gas: $115
Coffee: $40

March:
Groceries: $295
Gas: $125
Coffee: $44
Coffee shop: $28

April:
Groceries: $310
Gas: $141
Coffee: $24

Total coffee spending: $296
```

Arrow key to scroll down, enter to make new line. CTRL+S to save and CTRL+X to exit.

Let's say later in the day we remember that we edited a file, but forgot what, we can SEARCH for it like this.

> `find ./ -mtime 0` FIND anything that has been modified in the last 0 days

```
vboxuser@ExampleName:~/search_practice$ find ./ -mtime 0
./
./budget.txt
```

This is most of the ways you'll be finding your files. Next lab is for FILTERING the content INSIDE the files.